

PROJECT

Vegetation Condition Monitoring and Change Detection Using Landsat and Google Earth Engine

Project Type

- Remote Sensing
- Environmental GIS
- Vegetation Monitoring
- Land Cover Analysis
- Change Detection
- Raster Analytics

PROJECT OVERVIEW

This project demonstrates a complete remote sensing workflow for monitoring vegetation condition, assessing ecosystem health, and detecting long-term vegetation changes using Landsat imagery and Google Earth Engine.

The analysis integrates cloud-free summer satellite composites, NDVI modeling, vegetation condition classification, and temporal change detection to evaluate vegetation dynamics between 2015 and 2025 across Tehran.

By transforming raw Earth observation data into GIS-ready environmental indicators, the workflow provides a scalable framework for vegetation monitoring, ecological assessment, environmental reporting, and land management applications.

PROJECT OBJECTIVES

- Generate cloud-free Landsat composites using Google Earth Engine
- Derive NDVI-based vegetation indicators
- Assess spatial patterns of vegetation condition
- Classify vegetation health into meaningful ecological categories
- Detect vegetation gain and loss over a ten-year period
- Produce GIS-ready raster products for environmental analysis
- Support evidence-based environmental monitoring and ecosystem management

DATA SOURCES

Dataset	Purpose
Landsat 8 Collection 2 Level-2	Historical vegetation analysis
Landsat 9 Collection 2 Level-2	Recent vegetation monitoring
Administrative Boundaries	Study area definition
Google Earth Engine Data Catalog	Cloud-based image processing

TOOLS & TECHNOLOGIES

- Google Earth Engine (JavaScript API)
- ArcGIS Pro
- Remote Sensing
- Landsat 8/9
- Raster Analysis
- Environmental GIS
- Change Detection
- Cartographic Design

WORKFLOW

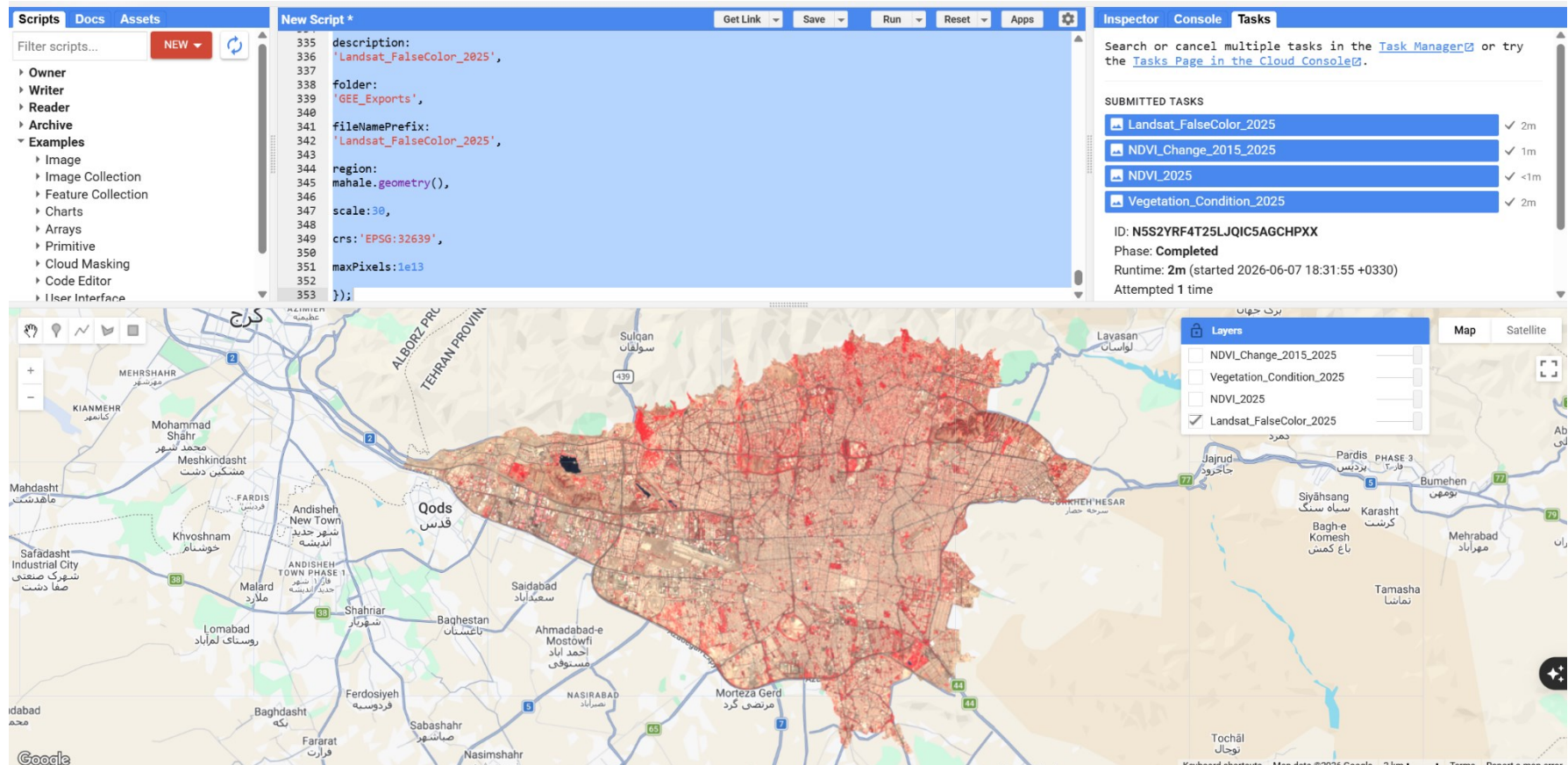
1. Satellite Data Preparation

Multi-temporal Landsat imagery was acquired and processed within Google Earth Engine.

The workflow included:

- Image filtering
- Summer-season selection (June–August)
- Cloud and shadow masking using QA_PIXEL
- Image compositing through median aggregation

The resulting cloud-free composites provide consistent seasonal observations for vegetation analysis.



2. NDVI Generation

Vegetation condition was quantified using the Normalized Difference Vegetation Index (NDVI).

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

NDVI values were calculated from Landsat Near-Infrared and Red spectral bands to measure vegetation vigor and photosynthetic activity. Higher values indicate healthier and denser vegetation cover, while lower values correspond to built-up, barren, or sparsely vegetated surfaces.

3. Vegetation Condition Classification

NDVI values were classified into ecological condition classes to facilitate interpretation and environmental reporting.

NDVI Range	Vegetation Condition
< 0.20	Sparse Vegetation
0.20 – 0.40	Moderate Vegetation
0.40 – 0.60	Healthy Vegetation
> 0.60	Dense Vegetation

For the study area, the observed NDVI range reached approximately 0.46, indicating that vegetation was primarily distributed within Sparse, Moderate, and Healthy vegetation categories, while Dense Vegetation was largely absent due to the urban character of the landscape.

4. Vegetation Change Detection

A multi-temporal change detection workflow was implemented by comparing NDVI conditions between 2015 and 2025.

The analysis quantified:

- Vegetation gain
- Vegetation loss
- Stable vegetation zones
- Spatial distribution of ecological change

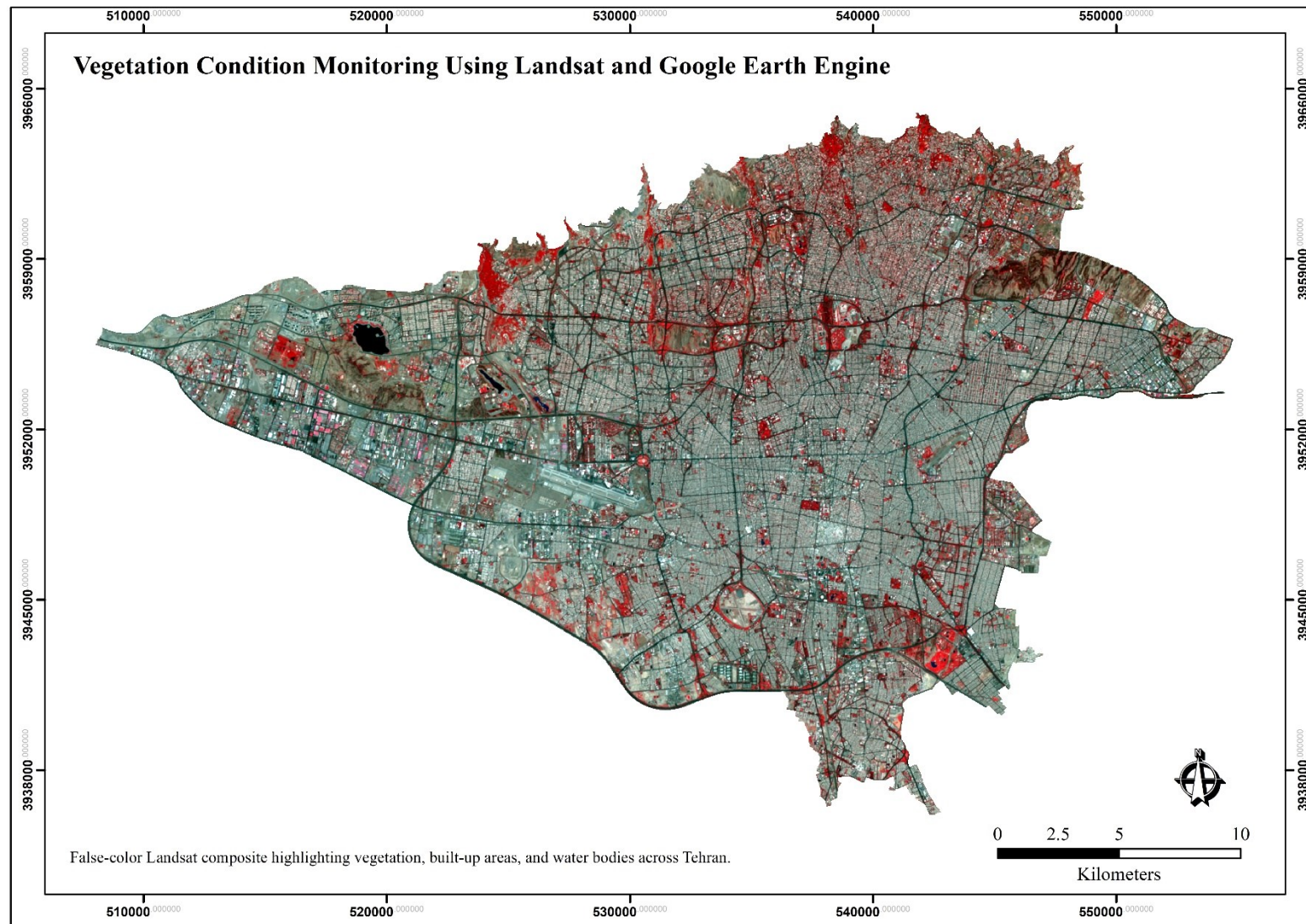
NDVI differencing enabled the identification of areas experiencing environmental improvement as well as locations potentially affected by vegetation degradation and urban expansion.

KEY OUTPUTS

Landsat False-Color Composite

A false-color Landsat composite was generated to visually distinguish vegetation, urban surfaces, and water features.

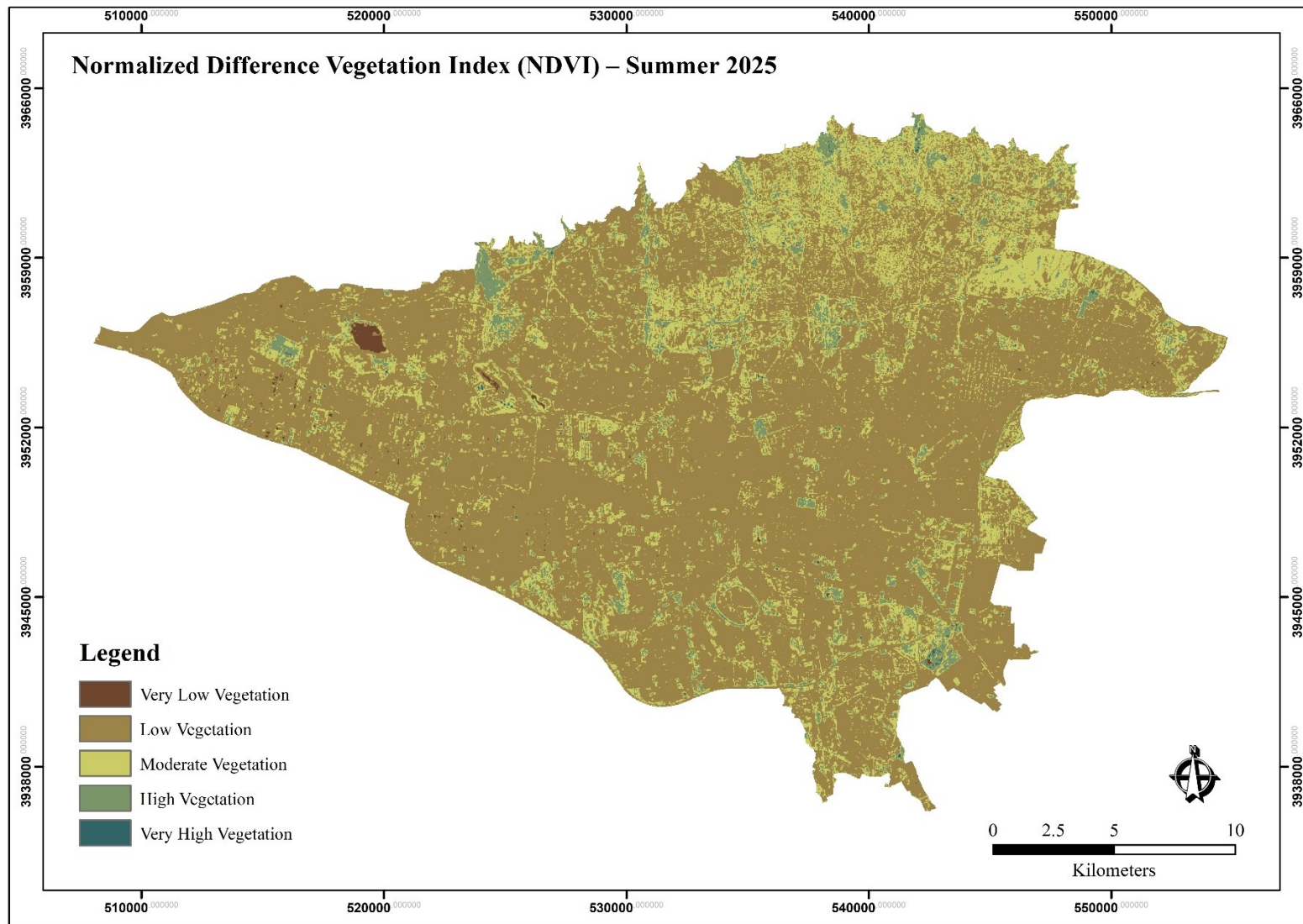
The composite provides direct visual verification of vegetation distribution and serves as the foundation for subsequent NDVI analysis.



NDVI Map

The NDVI map revealed clear spatial contrasts in vegetation condition across the metropolitan landscape.

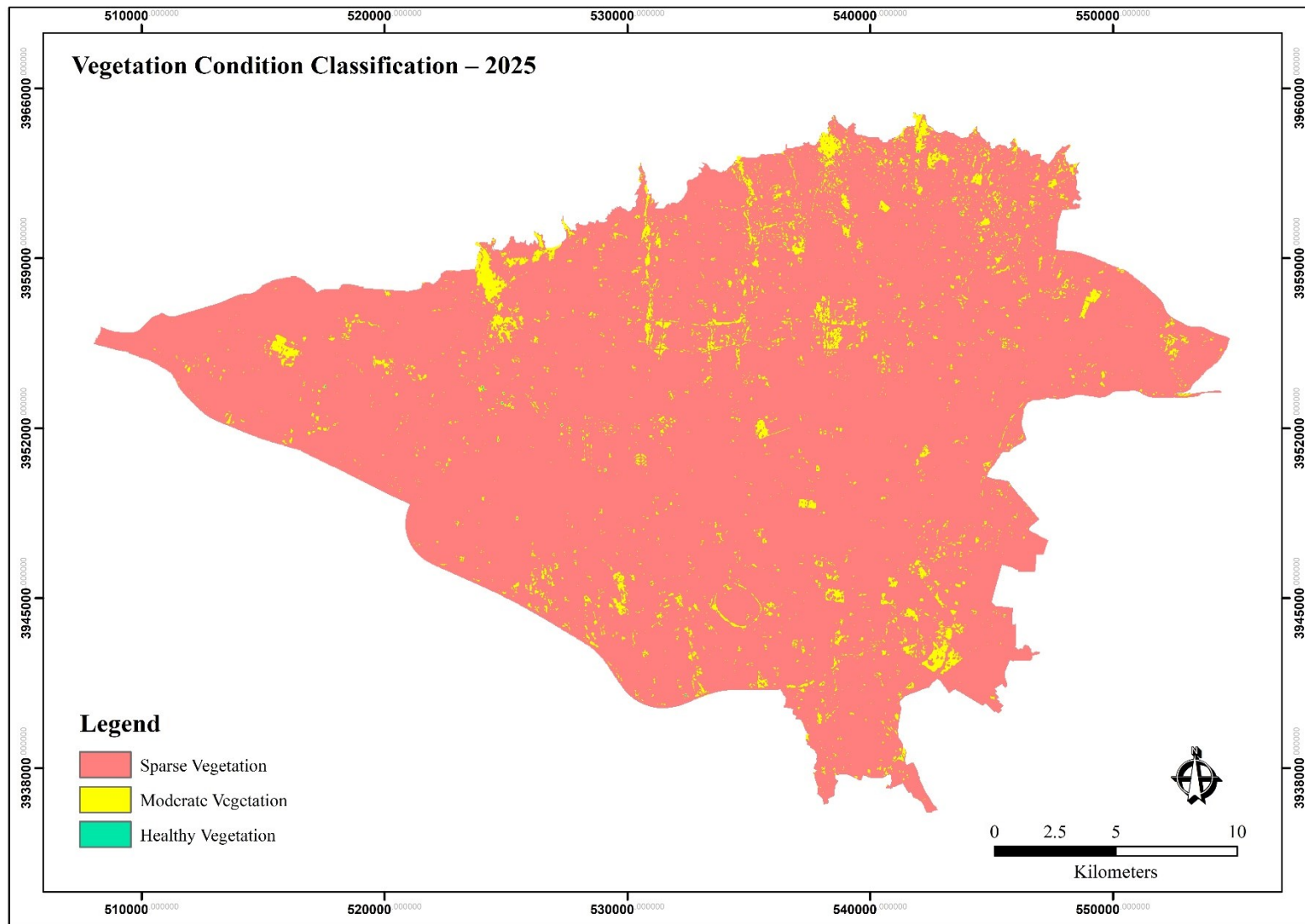
Higher NDVI values were concentrated within major parks, mountainous northern districts, and vegetated corridors, while dense urbanized areas exhibited substantially lower vegetation intensity.



Vegetation Condition Map

The classification results highlighted the predominance of sparse vegetation throughout the urban area, with moderate and healthy vegetation concentrated in localized green spaces and environmentally favorable zones.

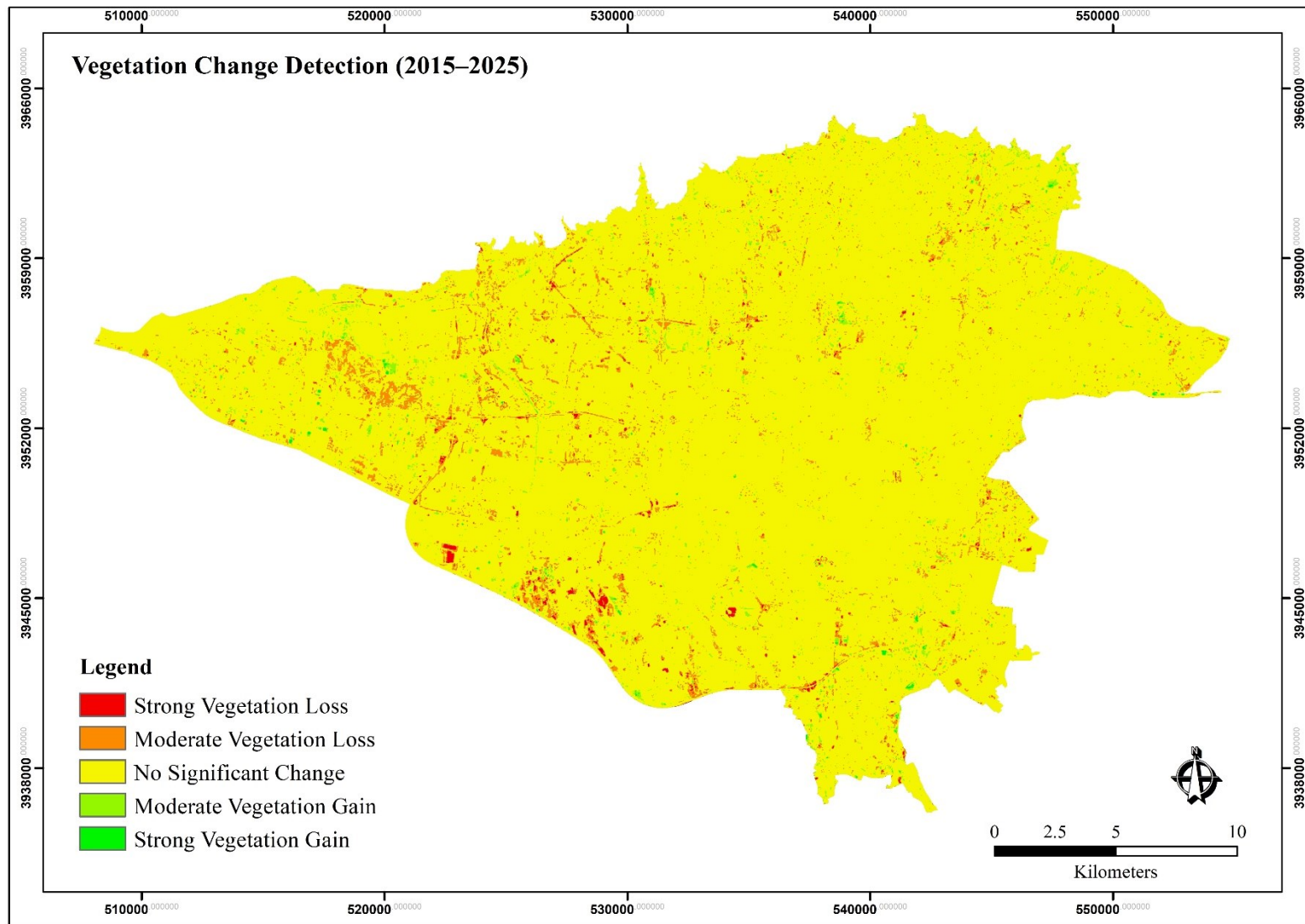
The map provides an interpretable representation of vegetation health suitable for environmental reporting and land management workflows.



Vegetation Change Map (2015–2025)

The change detection analysis identified areas of both vegetation gain and vegetation loss over the ten-year study period.

The results demonstrate how Earth observation data can be used to monitor long-term environmental dynamics and support ecosystem assessment initiatives.



KEY INSIGHTS

- NDVI successfully captured spatial variability in vegetation condition across the study area.
- Vegetation health was strongly concentrated within established green infrastructure and natural landscape zones.
- Urbanized districts exhibited substantially lower vegetation density.
- Multi-temporal NDVI analysis enabled the detection of long-term vegetation gain and loss patterns.
- Google Earth Engine provided an efficient framework for large-scale remote sensing processing and environmental monitoring.
- The workflow is transferable to vegetation mapping, land cover assessment, carbon projects, ecosystem monitoring, and environmental reporting applications.

MY ROLE

- Designed and implemented the complete Google Earth Engine workflow
- Processed and filtered Landsat imagery
- Developed cloud-masking procedures
- Generated NDVI and vegetation condition datasets
- Performed temporal change detection analysis
- Exported GIS-ready raster products
- Designed cartographic layouts and thematic maps
- Interpreted environmental patterns and vegetation dynamics

PROJECT VALUE

This project demonstrates practical expertise in remote sensing, vegetation mapping, raster analytics, and environmental monitoring using Google Earth Engine and GIS technologies.

The methodology is directly applicable to vegetation condition assessment, land cover mapping, carbon farming initiatives, ecological monitoring programs, environmental impact assessments, and ACCU-style environmental reporting workflows in Australia.